

NAAN MUDHALVAN

SERVICENOW PROJECT REPORT

PROJECT TITLE

Script-Controlled ACL - Restrict Record Access Based on Field Value

ServiceNow Administration Project

Submitted as part of the Naan Mudhalvan Skill Development Programme

Student Name: N. Abarnadevi

Register Number: C4S47501

Student Name:S.Abinaya

Register Number:C4S47502

Student Name:M.Abitha

Register Number:C4S47504

Student Name:P. Bhuvaneshwari

Register Number:C4S47505

Student Name:S. Devipriyalakshmi

Register Number:C4S47506

Course /Department: B.Sc Computer science

Institution: Government arts and science
college, thiruchuli

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CERTIFICATE

This is to certify that the project report entitled “Script-Controlled ACL – Restrict Record Access Based on Field Value” has been prepared as part of the Naan Mudhalvan ServiceNow Administration project work. The report covers the project idea, requirements, design, planning, development, testing, documentation and demonstration activities.

Project Guide / Faculty Signature: _____

Head of the Department: _____

Date: _____

DECLARATION

I hereby declare that this project report titled “Script-Controlled ACL – Restrict Record Access Based on Field Value” has been prepared by me for academic and skill-development purposes under the Naan Mudhalvan ServiceNow programme. The report presents the configuration concept, script logic, testing approach and documentation of the proposed access-control solution.

Student Signature: _____

Date: _____

ACKNOWLEDGEMENT

I sincerely thank the Naan Mudhalvan programme and the trainers who provided guidance and learning resources for ServiceNow administration. I also thank my faculty members and institution for providing the opportunity to complete this project. Finally, I thank everyone who supported me during the learning, configuration, testing and documentation stages of the project.

ABSTRACT

Access control is an important part of an enterprise application because users should only be able to view or modify records that they are authorized to access. This project demonstrates a field-value-based access-control concept in ServiceNow using a Script-Controlled Access Control List (ACL). The ACL script evaluates a selected field in the current record and decides whether the requested operation should be allowed. For example, a record may be accessible when an access-status field contains an approved value and restricted when the value is pending or rejected. The project covers the complete lifecycle from brainstorming and requirement analysis to design, planning, development, testing, documentation and demonstration.

Keywords: ServiceNow, ACL, Access Control, Field Value, Security, Script, Record Access, Authorization

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1. BRAINSTORMING & IDEATION PHASE

1.1 Project Idea

The project idea is to create a controlled record-access mechanism in ServiceNow. The solution uses an ACL script to inspect a value stored in a selected field of the current record. Based on the configured condition, the script returns an allow or deny decision. This provides an example of dynamic authorization rather than depending only on a fixed role.

1.2 Problem Identification

In a service-management system, different records may require different levels of access. A user can have a valid role and still should not necessarily view every record. A field such as access status, department, classification, approval state or ownership can be used as an additional condition. The project addresses this requirement by evaluating a record field during ACL processing.

1.3 Proposed Solution

A Script-Controlled ACL is created for the required table and operation, such as read. The script reads the relevant field from the current record and compares it with the permitted value. If the condition is satisfied, the ACL allows access; otherwise, access is denied, subject to other applicable security rules.

1.4 Expected Benefits

The approach demonstrates fine-grained authorization, clear security logic, easier testing of conditions and the ability to adapt access rules to business requirements. It also provides practical experience with ACL configuration and server-side scripting concepts.

2. REQUIREMENT ANALYSIS PHASE

2.1 Functional Requirements

ID	Requirement
FR1	User must be authenticated before requesting records.
FR2	The system must identify the requested record.
FR3	The ACL must read the configured field value.
FR4	The script must compare the field value with the access rule.
FR5	Access must be allowed when the condition is satisfied.
FR6	Access must be denied when the condition is not satisfied.
FR7	The ACL must be applied to the intended table and operation.

2.2 Non-Functional Requirements

Security: unauthorized records should be restricted. Reliability: the access rule should behave consistently. Maintainability: the script should be simple and understandable. Performance: the ACL should avoid unnecessary processing. Usability: the demonstration should make the access decision easy to verify.

2.3 Required Components

ServiceNow instance, target table, selected field, user/role configuration, ACL record, ACL script, test records, test users or roles, browser and documentation/screenshots.

2.4 Assumptions

The administrator has permission to create or modify ACLs. The selected field exists on the target table. Test records contain different field values so both allowed and denied scenarios can be demonstrated.

3. PROJECT DESIGN PHASE

3.1 High-Level Architecture

User Login → Record Request → ACL Evaluation → Read Field Value → Compare with Rule → Allow / Deny Access

3.2 Access-Control Flow

Step	Action	Result
1	User requests a record	Request reaches ACL layer
2	Applicable ACL is identified	Configured rule is evaluated
3	Script reads selected field	Current value is obtained
4	Value is compared	Condition becomes true or false
5	ACL returns decision	Access is allowed or denied

3.3 ACL Logic Design

The design separates the access decision into a simple condition. The selected field represents the record state that controls access. The script initializes the decision to false and changes it to true only when the configured condition is met.

3.4 Example Field

For demonstration, a field such as `u_access_status` can be used with values such as `Approved`, `Pending` and `Rejected`. The actual field name and value should be replaced with the field configured in the student's ServiceNow instance.

4. PROJECT PLANNING PHASE

4.1 Work Plan

Phase	Main Activity	Output
1	Study requirement	Problem statement and objective
2	Select table and field	Target configuration identified
3	Design ACL logic	Access flow and condition
4	Configure ACL	ACL record created
5	Develop script	Field-based access logic
6	Create test records	Allowed and restricted data
7	Execute testing	Test results and observations
8	Prepare report	Screenshots, results and conclusion
9	Demonstrate	End-to-end presentation

4.2 Risk and Mitigation

Incorrect field names can cause unexpected behavior; verify the dictionary name before testing. Testing only one record may hide errors; create multiple records with different values. Existing ACLs may also affect access; test the complete security path and review applicable ACLs. Use an appropriate training or development instance when available.

4.3 Deliverables

Configured ACL, working script, test data, test-case results, screenshots, project report and final demonstration.

5. PROJECT DEVELOPMENT PHASE

5.1 Environment Preparation

Open the ServiceNow instance used for the Naan Mudhalvan learning activity. Confirm that the required table, field and user/role setup are available. Identify the exact dictionary name of the field that will control access.

5.2 ACL Configuration

Create or configure an ACL for the required table and operation, for example the read operation. Select the appropriate role or condition as required by the project. Enable the script section so that the field-value rule can be evaluated.

5.3 Sample Script

The following is an illustrative server-side ACL script. Replace the field name and permitted value with the values configured in the actual instance:

```
answer = false;

if (current.u_access_status == 'Approved') {
    answer = true;
}
```

5.4 Script Explanation

The variable answer represents the ACL decision. It starts as false, so the default is deny. The condition checks the current record's access-status field. When the value is Approved, the decision becomes true. For other values, the script keeps the default deny decision.

5.5 Development Checklist

Verify table → verify operation → verify field name → configure ACL → add script → save → test approved record → test non-approved record → review other applicable ACLs → document results.

6. PROJECT TESTING PHASE

6.1 Testing Objective

Testing verifies that the ACL permits access only when the configured field-value condition is satisfied and denies access when it is not satisfied.

6.2 Test Cases

TC	Field Value	Expected Result	Status
TC01	Approved	Access allowed	Pass / Fail
TC02	Pending	Access denied	Pass / Fail
TC03	Rejected	Access denied	Pass / Fail
TC04	Blank / unexpected	Access denied	Pass / Fail
TC05	Approved but role requirement not met	Access denied	Pass / Fail

6.3 Testing Procedure

1. Log in with the test user. 2. Open the target table. 3. Select a record whose field contains Approved. 4. Confirm access when all ACL requirements are satisfied. 5. Repeat with Pending and Rejected records. 6. Observe the access behavior. 7. Record screenshots and results. 8. Review ACL configuration if behavior differs from the expected result.

6.4 Expected Testing Outcome

The test should demonstrate a clear difference between records that satisfy the field-value condition and records that do not. The exact result depends on the complete set of ACLs and roles configured in the instance.

7. PROJECT DOCUMENTATION PHASE

7.1 Documentation Structure

The final documentation contains the project title, certificate, declaration, acknowledgement, abstract, objectives, eight project phases, screenshots, script explanation, testing table, conclusion and references.

7.2 Screenshots to Include

Recommended screenshots are: ServiceNow instance page; target table and field; ACL configuration page; ACL script section; sample Approved record; sample Pending/Rejected record; successful access result; restricted-access result; and final project demonstration screen.

7.3 Security Considerations

Access-control scripts should follow the principle of least privilege. The rule should be limited to the intended table and operation. Avoid exposing sensitive information through debugging output. Test changes before applying them to production environments. Keep the logic simple and maintainable.

7.4 Maintenance

If the business requirement changes, update the permitted field values or access logic and repeat the test cases. Document significant changes so future users can understand why the ACL was configured.

8. PROJECT DEMONSTRATION PHASE

8.1 Demonstration Sequence

Step 1: Introduce the project title and objective. Step 2: Show the selected table and access-control field. Step 3: Open the ACL configuration. Step 4: Explain the script. Step 5: Open a record with the permitted field value. Step 6: Demonstrate allowed access. Step 7: Open a record with a non-permitted value. Step 8: Demonstrate restricted access. Step 9: Show the test-case table. Step 10: Summarize the project and conclusion.

8.2 Sample Presentation Explanation

“This project implements a Script-Controlled ACL in ServiceNow. The ACL checks a selected field value before allowing access to a record. When the value satisfies the configured condition, the script returns true and access is permitted, subject to other applicable ACL requirements. When the condition is not satisfied, the default decision remains false and access is restricted. The testing phase verifies both allowed and denied scenarios.”

8.3 Demonstration Checklist

- ☐ Project title explained
- ☐ Problem explained
- ☐ Requirements shown
- ☐ ACL configuration shown
- ☐ Script explained
- ☐ Allowed record tested
- ☐ Restricted record tested
- ☐ Test results shown
- ☐ Conclusion explained

9. PROJECT OBJECTIVE

The main objective of this project is to demonstrate how a ServiceNow ACL script can restrict record access according to a field value. The project aims to provide practical understanding of dynamic authorization, ACL configuration, server-side scripting, security testing and documentation.

10. CONCLUSION

The project demonstrates a field-based access-control approach using a Script-Controlled ACL in ServiceNow. The complete workflow includes identifying the access problem, analyzing requirements, designing the rule, planning configuration activities, developing the ACL script, testing different field values, documenting the implementation and demonstrating the final behavior. The project provides practical exposure to ServiceNow security concepts and shows how record access can be controlled using business-specific conditions.

11. REFERENCES / LEARNING RESOURCES

1. ServiceNow platform learning materials used in the Naan Mudhalvan training programme.
2. ServiceNow Administration and Access Control List learning resources available through the training instance.
3. Project notes, configuration observations and test results prepared during the project activity.